

Model Discrimination in the DES-SN5YR Hubble Diagram Is Highly Sensitive to the Simulation-Derived Bias-Correction Layer

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Code and reproduction package: github.com/tracyphasespace/Model-Discrimination-DES-SN5YR

Abstract

Published Type Ia supernova distance moduli are not raw measurements. They are the output of a processing chain that includes light-curve standardization, Galactic-extinction treatment, photometric calibration, redshift and peculiar-velocity corrections, and survey-bias corrections derived from simulations. In the DES-SN5YR release, the BEAMS with Bias Corrections (BBC) layer is obtained from simulations generated under a *reference* cosmology together with intrinsic-scatter and population models.

We quantify the leverage of that correction step in the public DES-SN5YR v1.2 Hubble-diagram release, using the official likelihood’s redshift convention (z_{HEL} in the external luminosity-distance factor, z_{HD} in the cosmological integral). The released BBC correction has median -0.05 mag, range -0.89 to $+0.65$ mag, and a redshift regression slope of -0.125 mag per unit z . Comparing flat Λ CDM against a one-parameter non-expanding probe law with the full released covariance and an analytically marginalized offset, the released vector gives $\Delta\chi^2 \equiv \chi_B^2 - \chi_A^2 = +1.1$, weakly favouring Λ CDM; the reconstructed pre-BBC vector (`MU + biasCor_mu`) gives $+12.8$. Applying the released BBC step therefore *changes* the pairwise statistic by $L_{\text{BBC}} = -11.7$ χ^2 units — roughly eleven times the final discrimination — and moves this comparison toward Model B rather than toward the flat- Λ CDM comparator, so the concern raised here does not arise from a correction that disadvantages the alternative. The result is unchanged when all 1829 released entries are retained.

The bracket measures the *total* leverage of the BBC step: removing the correction simultaneously restores real survey selection (χ^2/dof degrades from 0.93 to 1.34), and the reconstruction retains the globally fitted standardization parameters, making it a hybrid diagnostic. Neither the magnitude nor the sign of any specifically reference-cosmology-conditioned component is isolated by the bracket. Crucially, DES has already tested that component directly: Camilleri et al. (2024) rerun the bias-correction simulations under reference cosmologies spanning distance-modulus variations of ~ 0.15 mag and find the effect subdominant to statistical uncertainties. We compute that the released-best-fit probe law differs in offset-removed shape from Λ CDM by at most 0.055 mag over the sample — well below that tested amplitude (an amplitude comparison; Camilleri’s formal criterion is the shape envelope bounded by their Fig. 4e reference curves, and the pointwise containment test against those boundary cosmologies is the direct completion). The finding is therefore sharper than a missing-validation claim: *the total BBC layer carries pairwise leverage an order of magnitude larger than the verdict even though prior DES simulation tests indicate the reference-cosmology part of that leverage is small at the amplitude scale of this comparison*. The untested regime begins at larger departures — our Einstein–de Sitter control

(0.276 mag, $1.8\times$ the tested amplitude) exhibits $|L_{\text{BBC}}| \approx 365$ — and for functional forms outside the FLRW family the tested coverage is an inference to be completed, not assumed. We report covariance-weighted projection and fixed/adaptive diagnostics, the verdict sensitivity to twelve released BBC-related systematic modes (0.71 in quadrature), and propose matched-simulation reruns and paired corrections as the completion of the validation programme. All inputs are public; a complete reproduction recipe is given in §7.

1 Introduction

Type Ia supernovae provide one of the principal observational constraints on the cosmic distance–redshift relation. Modern surveys do not compare cosmological models directly with raw photon counts; they construct standardized distance estimates through a processing chain incorporating instrumental calibration, light-curve fitting, standardization relations, redshift corrections, population assumptions, contamination treatment, and corrections for survey selection. This distinction matters whenever the statistical separation between competing distance laws is small.

The DES-SN5YR cosmological sample (DES Collaboration, 2024; Vincenzi et al., 2024; Sánchez et al., 2024) is sufficiently precise that differences between plausible fitted models can be of order a few units in χ^2 . At that scale a processing component need not be large in absolute magnitude to affect model discrimination; it need only project onto the difference between the distance laws being compared.

This paper examines one such component: the BEAMS with Bias Corrections framework (Kessler & Scolnic, 2017) used in the DES-SN5YR analysis. BBC uses survey simulations to estimate selection and fitting biases and applies an event-level correction to the standardized distances. Two distinct cosmology inputs appear in that machinery and must not be conflated (Camilleri et al., 2024): the *reference* cosmology used to generate the bias-correction simulations, and the *fiducial* cosmology used internally while determining the standardization nuisance parameters and redshift-bin offsets. Throughout this paper “conditioning” refers to dependence on the reference cosmology unless stated otherwise.

The existence of this conditioning is not itself a defect, and it is not untested. Kessler & Scolnic (2017) varied the simulation cosmology within the conventional w CDM neighbourhood and found small effects on recovered parameters. More directly, Camilleri et al. (2024) rerun the DES-SN5YR bias-correction simulations under reference cosmologies differing from the baseline by up to $\sim 10\sigma$ — distance-modulus variations of roughly 0.15 mag — including nonstandard w_0w_a truth models, and find the impact subdominant to statistical uncertainties, providing an iterative prescription for cases beyond. The open questions are therefore quantitative and specific: how large is the *total* BBC step’s leverage on a pairwise comparison; where does a given alternative law sit relative to the validated reference-cosmology envelope; and what remains for laws outside it or outside the FLRW functional family. Those are the questions measured here. This paper proposes no replacement cosmology; the alternative law is a probe, and readers may substitute any smooth one-parameter alternative.

2 Data and provenance

We use the public DES-SN5YR Hubble-diagram release, tag v1.2 (DES Collaboration, 2024; Sánchez et al., 2024). The relevant files are `DES-SN5YR_HD+MetaData.csv` (containing z_{HD} , z_{HEL} , the released modulus `MU`, its uncertainty `MUERR_FINAL`, `biasCor_mu`, SALT parameters `mB`, `x1`, `c`, survey identifiers, and peculiar-velocity information) and the released systematic covariance. Per the

release documentation, `STAT+SYS.txt.gz` contains systematics only; the statistical term is carried in `MUERR_FINAL` and the total covariance is $C = C_{\text{sys}} + \text{diag}(\text{MUERR_FINAL}^2)$.

The v1.2 release contains 1829 Hubble-diagram entries. Our default sample excludes 61 effectively zero-weight entries with `MUERR_FINAL` > 10, leaving $N = 1768$ over $0.025 \leq z_{\text{HD}} \leq 1.121$ (1574 DES-discovered supernovae plus 194 external low-redshift objects); this is our diagnostic cut, not the official likelihood convention, which retains all 1829. As a robustness check, retaining all 1829 entries changes no reported number at the quoted precision. The release version is frozen because the DES-SN pipeline has subsequently evolved.

Distance moduli are compared with model predictions using the official likelihood’s redshift convention: the cosmological integral is evaluated at z_{HD} while the external luminosity-distance factor uses the heliocentric redshift, $\mu \propto 5 \log_{10}[(1 + z_{\text{HEL}}) D_M(z_{\text{HD}})]$. The probe law adopts the analogous deliberate assignment (§5.2).

3 The processing layers

3.1 Framework-neutral or shared layers

Photometric calibration and Galactic-extinction treatment are instrumental or astrophysical corrections rather than assumptions about the cosmological distance–redshift relation. The SALT light-curve fit (Kenworthy et al., 2021) also transforms the observed phase coordinate by the measured time-dilation relation; DES light curves independently show an approximately $(1 + z)$ temporal broadening, $b = 1.003 \pm 0.005$ (stat) ± 0.010 (sys) (White et al., 2024). Any framework tested against these standardized data must reproduce that observation, so the present experiment treats the phase transformation as shared processing.

3.2 The BBC bias correction

The layer investigated directly is `biasCor_mu`. DES documents the released distance as

$$\mu = -2.5 \log_{10} x_0 + \alpha x_1 - \beta c \pm \gamma/2 - \text{biasCor}_\mu - M_0, \quad (1)$$

with global α , β , γ determined jointly with the bias-corrected Hubble diagram. For the $N = 1768$ sample the released correction has median -0.05 mag; range -0.89 to $+0.65$ mag; binned medians, low to high z : $(-0.053, -0.025, -0.069, -0.072, -0.139, -0.118)$ mag; and a linear regression slope of -0.125 mag per unit z . The binned relation is not monotonic; the regression slope is a compact description of the systematic redshift dependence, not a claim that BBC follows a linear law. The relevance of the correction is set not by amplitude but by its projection onto the model difference, measured in §5.7.

3.3 Peculiar velocities

The 194 external low-redshift objects carry peculiar-velocity corrections (median $|v| \approx 121 \text{ km s}^{-1}$) from flow reconstructions (Peterson et al., 2022). We do not treat these as evidence for the central BBC claim. An anchor-exclusion variant is reported in §5.6 as a sample-composition sensitivity test only.

4 Statistical question: discrimination leverage

Let $\Delta\chi^2 \equiv \chi_B^2 - \chi_A^2$, with Model A flat Λ CDM and Model B the probe; positive values favour Model A. For a distance vector $\boldsymbol{\mu}$, model prediction $\boldsymbol{m}$, and covariance C , we minimize $\chi^2 = \boldsymbol{r}^T C^{-1} \boldsymbol{r}$ with $\boldsymbol{r} = \boldsymbol{\mu} - \boldsymbol{m}$ after analytic marginalization over an additive magnitude offset.

The pairwise BBC discrimination leverage is

$$L_{\text{BBC}} \equiv \Delta\chi_{\text{released}}^2 - \Delta\chi_{\text{preBBC}}^2. \quad (2)$$

Negative L_{BBC} means that applying BBC shifts the comparison toward Model B relative to the reconstructed pre-BBC endpoint. The statement applies to the total correction only.

Amplitude alone is not sufficient: a large correction can be irrelevant if covariance-orthogonal to the model difference, and a small one decisive if aligned. With the offset direction projected out (tildes), the covariance-weighted cosine is

$$\rho_C = \frac{\tilde{\boldsymbol{b}}^T C^{-1} \tilde{\boldsymbol{d}}}{\sqrt{(\tilde{\boldsymbol{b}}^T C^{-1} \tilde{\boldsymbol{b}})(\tilde{\boldsymbol{d}}^T C^{-1} \tilde{\boldsymbol{d}})}}, \quad (3)$$

where $\boldsymbol{b}$ is the BBC vector and $\boldsymbol{d} = \boldsymbol{m}_B - \boldsymbol{m}_A$ the model-separation vector.

5 The BBC bracket

5.1 Empirical determination of the sign convention

The sign of `biasCor_mu` must be established before reconstructing a pre-BBC vector. We determine it from the data: regressing $(\mu_{\text{target}} - m_B)$ on $(x_1, -c, 1)$ for the three candidate conventions, only `MU + biasCor_mu` reconstructs a Tripp relation (Tripp, 1998) with physically standard coefficients. The unconstrained regression returns $\alpha = 0.15546 \pm 0.00038$ and $\beta = 3.1159 \pm 0.0037$ with rms residual 0.0165 mag — near the DES global values $\alpha = 0.16087$, $\beta = 3.11780$ (Vincenzi et al., 2024); the residual offset in α and the rms floor reflect the omission of the host-mass step γ and of per-event weighting from this diagnostic regression. The released modulus therefore contains the correction with a minus sign, per Eq. (1), and the reconstructed pre-BBC vector is

$$\mu_{\text{preBBC}} = \text{MU} + \text{biasCor_mu}. \quad (4)$$

An earlier draft of this analysis used the opposite convention, thereby applying the correction twice, and reported a spurious verdict reversal; the error was caught by this check before circulation.

5.2 Distance laws

Model A (flat Λ CDM). The standard spatially flat luminosity distance evaluated per the official convention, $\mu_A \propto 5 \log_{10}[(1 + z_{\text{HEL}}) D_M(z_{\text{HD}})]$, with one fitted shape parameter Ω_m ; the absolute magnitude/ H_0 combination is degenerate with the additive offset and is analytically marginalized.

Model B (non-expanding phenomenological probe).

$$D(z_{\text{HD}}) = \frac{c}{K} \ln(1 + z_{\text{HD}}), \quad \mu_B \propto 5 \log_{10}[(1 + z_{\text{HEL}}) D(z_{\text{HD}})] + \Delta\mu, \quad (5)$$

with attenuation

$$\Delta\mu(z_{\text{HD}}) = \frac{5}{\ln 10} \eta \tau(z_{\text{HD}}), \quad \tau(z) = 1 - (1 + z)^{-1/2}, \quad (6)$$

one fitted shape parameter η , and K marginalized with the offset. The redshift assignment is deliberate: observed-frame flux factors carry z_{HEL} ; the accumulated path and attenuation coordinate is z_{HD} . Model B supplies a smooth distance–redshift relation substantially different in functional form from the baseline FLRW relation; no physical claim is attached to it here.

5.3 Headline results

Table 1: The sensitivity bracket ($N = 1768$, full STAT+SYS covariance, official redshift convention, offset marginalized). Positive $\Delta\chi^2$ favours flat Λ CDM. Retaining all 1829 released entries changes no entry at this precision.

Vector	Model A	Model B	$\Delta\chi^2$	χ^2/dof
Released (BBC-corrected)	$\Omega_m = 0.352, \chi^2 = 1640.1$	$\eta = 0.297, \chi^2 = 1641.1$	+1.1	0.93
Reconstructed pre-BBC	$\Omega_m = 0.499, \chi^2 = 2361.8$	$\eta = 0.049, \chi^2 = 2374.5$	+12.8	1.34

There is no reversal of the pairwise ordering. Applying the released BBC step *changes* the statistic from +12.8 at the reconstructed endpoint to +1.1 in the released vector:

$$L_{\text{BBC}} = 1.1 - 12.8 = -11.7, \quad (7)$$

a leverage of roughly eleven times the released discrimination, directed toward Model B. In this comparison the total correction moves the statistic toward the alternative probe rather than toward the flat- Λ CDM comparator; the methodological concern raised here therefore does not arise from objecting to a correction because it disadvantages Model B. The fitted shapes are strongly sensitive to the step ($\Omega_m : 0.499 \rightarrow 0.352$; $\eta : 0.049 \rightarrow 0.297$), which establishes sensitivity, not that BBC “supplies” either model’s structure.

5.4 What the bracket establishes, and what it does not

The bracket establishes: (i) the BBC step has substantial covariance-weighted leverage on this comparison; (ii) that leverage is an order of magnitude larger than the statistic used to discriminate the models; (iii) the direction of the *total* correction is known. It does not establish: that reference-cosmology conditioning creates the released near-degeneracy; the sign of the conditioned component; that the pre-BBC vector is closer to the true Hubble diagram; that Model B is physically correct; or that every alternative law exhibits the same sensitivity (tested in §5.6).

Three structural caveats govern the interpretation. First, removing the correction restores genuine selection effects — the degradation $\chi^2/\text{dof} : 0.93 \rightarrow 1.34$ is the tell that both models are then fitting a z -dependent selection ramp neither describes — and the released covariance is not the covariance of the reconstructed vector, so only differences within a row of Table 1 are quoted. Second, the reconstructed pre-BBC vector is a *hybrid diagnostic* rather than the result of rerunning the analysis without BBC: adding `biasCor_mu` removes the released event-level magnitude correction but leaves in place the globally determined standardization parameters α , β , γ , together with other quantities obtained within the nominal analysis; the direction and magnitude of this residual nuisance-parameter conditioning are not determined by the present bracket. Third, writing the measured total schematically as

$$L_{\text{BBC}} = L_{\text{selection}} + L_{\text{population}} + L_{\text{scatter}} + L_{\text{classification}} + L_{\text{cosmology}} + L_{\text{interactions}}, \quad (8)$$

the experiment measures the left-hand side only; the components can add, cancel, and interact, so no term is individually bounded by the total.

5.5 How small must the conditioned component be?

The released comparison is decided by $\Delta\chi^2 = 1.1$; the total correction-step swing is 11.7. For the reference-cosmology contribution merely to remain smaller than the nominal discrimination requires $|L_{\text{cosmology}}| < 1.1$, i.e. below roughly 9% of the observed total swing — and considerably smaller to be negligible. §6.1 addresses how much of that requirement prior DES validation already discharges for this probe.

5.6 Robustness and second probes

Table 2: Pairwise discrimination leverage for three distance-law probes and two analysis variants (official redshift convention). Positive $\Delta\chi^2$ favours flat Λ CDM (free Ω_m).

Probe / variant	$\Delta\chi^2_{\text{released}}$	$\Delta\chi^2_{\text{preBBC}}$	L_{BBC}
Model B, non-expanding (headline)	+1.1	+12.8	−11.7
Flat w CDM, $w = -0.5$, Ω_m fitted	+2.9	+8.6	−5.7
Einstein–de Sitter ($\Omega_m = 1$, no free shape)	+754.7	+389.4	+365.3
Statistical-only covariance (Model B)	+0.4	+30.0	−29.6
DES-only, external anchors excluded (Model B)	+6.5	+9.9	−3.4

BBC leverage is substantial across the tested comparisons; it exceeds the released discrimination in the headline Model-B comparison, the $w = -0.5$ probe, and the statistical-only variant, while for the Einstein–de Sitter control and the anchor-excluded variant the leverage, though large in absolute terms, is smaller than those comparisons’ own (much larger) released margins. The sensitivity is not unique to Model B: substantial leverage is also observed for two additional separated distance-law probes. The *sign* of the leverage varies by probe (+365, −11.7, −5.7): the correction moved the Einstein–de Sitter comparison in Λ CDM’s favour and the Model-B comparison toward Model B, so the layer displays no consistent partisanship; its direction is set by the geometry of each comparison. The Einstein–de Sitter case is a sensitivity control only (its absolute χ^2 is catastrophic on either vector, as expected). The statistical-only variant shows that covariance treatment changes the numerical leverage materially and cannot replace the full covariance. The anchor-exclusion variant is a sample-composition test: it removes redshift coverage and the normalization–shape lever arm along with the peculiar-velocity layer, so no attribution to the latter is made.

5.7 Covariance-weighted projection

Evaluating Eq. (3) with $\mathbf{d}$ at the tabulated three-decimal released-vector best-fit values ($\Omega_m = 0.352$, $\eta = 0.297$; evaluating instead at full-precision optimizer outputs shifts L^{fixed} by ~ 0.2):

$$\rho_C = -0.068, \quad \|\tilde{\mathbf{b}}\|_{C^{-1}} = 29.76, \quad \|\tilde{\mathbf{d}}\|_{C^{-1}} = 2.51. \quad (9)$$

Repeating with $\mathbf{d}$ at the pre-BBC best fits gives $\rho_C = -0.077$: the diagnostic is stable against the evaluation point.

The covariance-weighted cosine is small — the vectors are nearly covariance-orthogonal, as an honest selection correction should be; the layer is not “aimed” at the comparison — yet the BBC vector’s norm in the covariance metric is large enough that its residual projection alone dominates the released margin (§5.8). The general requirement on the conditioned component follows from

$$2|\tilde{\mathbf{b}}_{\text{cond}}^T C^{-1} \tilde{\mathbf{d}}| < 1.1:$$

$$|\rho_{\text{cond}}| \|\tilde{\mathbf{b}}_{\text{cond}}\|_{C^{-1}} < \frac{1.1}{2\|\tilde{\mathbf{d}}\|_{C^{-1}}} \simeq 0.22, \quad (10)$$

a joint bound on the conditioned component’s norm and alignment; no assumption about its norm is made.

5.8 Fixed-shape versus adaptive leverage

Holding both models’ shape parameters at the released-vector best fits and recomputing the bracket separates the direct effect of the correction from parameter accommodation:

$$\begin{aligned} L_{\text{BBC}}^{\text{fixed}} &= 2\tilde{\mathbf{b}}^T C^{-1} \tilde{\mathbf{d}} = 2\rho_C \|\tilde{\mathbf{b}}\|_{C^{-1}} \|\tilde{\mathbf{d}}\|_{C^{-1}} \\ &\simeq 2(-0.068)(29.76)(2.51) = -10.1, \end{aligned} \quad (11)$$

verified independently by direct evaluation of the fixed-shape bracket. The remainder,

$$L_{\text{BBC}}^{\text{adapt}} = L_{\text{BBC}} - L_{\text{BBC}}^{\text{fixed}} = -1.6, \quad (12)$$

is the contribution of parameter accommodation. Most of the leverage is direct alignment of the correction with the model difference, and the agreement between the projection identity and the independently evaluated fixed-shape bracket is an internal consistency check on the computation.

6 The reference-cosmology component

6.1 Where the probe sits relative to the tested amplitude scale

Camilleri et al. (2024) distinguish the reference cosmology (generating the bias-correction simulations) from the fiducial cosmology (used inside the BBC nuisance fit) and test both dependencies directly, rerunning the simulation chain under reference cosmologies as far as $\sim 10\sigma$ from the baseline — distance-modulus variations of roughly 0.15 mag, including nonstandard $w_0 w_a$ truth models — and find the effect on the recovered cosmology subdominant to DES statistical uncertainties.

We compute where the present comparison sits relative to the amplitude scale of those tests. After covariance-weighted removal of the free offset, the released-best-fit Model B differs from flat Λ CDM by at most 0.055 mag over the sample (0.035 mag against $\Omega_m = 0.315$), and the pre-BBC-best-fit curve by at most 0.053 mag — substantially below the ~ 0.15 mag reference-cosmology variation Camilleri et al. explicitly tested. By contrast the Einstein–de Sitter control differs by 0.276 mag, a factor 1.8 *above* that tested amplitude, and is precisely the probe exhibiting two orders of magnitude larger leverage.

This amplitude comparison places the headline probe well within the scale of the Camilleri tests, and the conclusion accordingly cuts in the incumbent’s favour for the headline comparison: prior DES simulation tests indicate the reference-cosmology part of the measured leverage is small at this amplitude. Two caveats define what remains: Camilleri’s formal validation criterion is the shape envelope bounded by the reference curves of their Fig. 4e, so the present scalar comparison is an amplitude argument rather than the pointwise shape-containment test (reconstructing those boundary cosmologies and testing containment directly is the named upgrade); and the envelope was established with FLRW-family reference simulations, so its coverage of non-FLRW functional forms (such as Model B’s logarithmic distance plus saturating attenuation) is inferential rather than direct. The direct completion for both cases is a matched rerun of the bias-correction simulations

with the alternative law as the reference input, comparing $L_{\text{cosmology}} = \Delta\chi^2(\mathbf{b}_B) - \Delta\chi^2(\mathbf{b}_A)$ against the released margin; Camilleri et al. (2024) provide the iterative prescription such a test would follow.

6.2 Sensitivity to the released BBC-related systematic modes

The release provides single-systematic covariance blocks for BBC-related inputs: the intrinsic-scatter model (BS20), dust/population realizations (P21SYS1–3, P21_HOSTCOLOR, W22_AGE), host library (SVAHOSTLIB), selection efficiency (HOSTEFFshift), core-collapse likelihood (CClikelihood), and classifier swaps (SNNtraining, SNIRF, SCONE). None of these is a reference-cosmology variation, so this test does *not* isolate the conditioned component; it measures the verdict movement carried by the released systematic budget these modes represent. Each block is (near) rank-1, $C_k \approx \mathbf{d}_k \mathbf{d}_k^T$; perturbing the released moduli by $\pm \mathbf{d}_k$ and refitting both models yields per-input verdict shifts between ± 0.01 and $\pm 0.40 \chi^2$ units, with quadrature sum 0.71 against the released margin of 1.1. The quadrature sensitivity of the pairwise verdict to these twelve selected modes is therefore below the margin — the declared budget they represent does not by itself overturn the verdict — while the reference-cosmology question is addressed by the simulation reruns of §6.1, not by this decomposition.

6.3 Paired corrections, and the definitive instrument

A useful intermediate is a paired-correction experiment: one common, empirically anchored survey-selection model; corrections computed under each candidate law; the verdict reported together with its spread across the two schemes. If the spread is comparable to the claimed discrimination, corrected-modulus comparison has not produced a conditioning-independent verdict. The definitive instrument is likelihood-level forward modelling: each candidate law propagated through the same population, intrinsic-scatter, cadence, detection, classification, and cut architecture, compared with the survey at the likelihood level. Selection is a generative problem; simulation should not be eliminated but shared.

7 Reproduction

The analysis requires the public v1.2 release and a scientific Python environment; the reference implementation is at github.com/tracyphasespace/Model-Discrimination-DES-SN5YR (seven scripts; each regenerates the numbers of the section named).

1. *Freeze the data.* Fetch `DES-SN5YR_HD+MetaData.csv` and `STAT+SYS.txt.gz` at tag v1.2 (`data/fetch.sh`); the twelve single-systematic blocks via `data/fetch_systematics.sh`; record hashes.
2. *Load and clean.* Default cut `MUERR_FINAL ≤ 10` gives $N = 1768$; the all-1829 variant is one environment flag. Build $C = C_{\text{sys}} + \text{diag}(\text{MUERR_FINAL}^2)$ (positive-definite; C_{sys} alone is not).
3. *Establish the sign empirically* (`sign_check.py`; §5.1). Do not proceed until this passes.
4. *Characterize the layer* without reference to Model B: median, extrema, binned medians, regression slope.
5. *Implement both laws* exactly as §5.2, including the $z_{\text{HD}}/z_{\text{HEL}}$ convention; marginalize the offset analytically.

6. *Fit and bracket* (`bracket.py`); confirm Table 1 and $L_{\text{BBC}} = -11.7$.
7. *Projection and decomposition* (`projection.py`); confirm Eqs. (9)–(12) including the identity check.
8. *Robustness and probes* (`robustness.py`, `probes.py`); confirm Table 2.
9. *Amplitude comparison* (`envelope.py`); confirm the §6.1 placements.
10. *Systematic modes* (`bbc_systematic_modes.py`); confirm the 0.71 quadrature.

Falsification criterion. The central claim fails if an independent implementation with the verified sign convention, frozen v1.2 data, the official redshift convention, matched covariance ordering, full covariance, and correct offset marginalization produces a correction-step swing that is not large relative to the released discrimination. We invite that check specifically. The first two implementation errors found by this procedure were the authors’ own (a sign-convention error and a redshift-convention simplification), both corrected before freezing; that is why the procedure is included.

8 Discussion

The result should not be read as an indictment of simulation-based bias correction. A flux-limited survey requires a model of the objects it fails to observe, and the pre-BBC endpoint demonstrates the point as clearly as it demonstrates sensitivity: the goodness of fit deteriorates substantially when the correction is backed out. Nor should it be read as a claim that DES has not tested cosmology dependence: Camilleri et al. (2024) tested the reference-cosmology dependence directly over a substantial amplitude range and found it small there, and the headline comparison of this paper sits well within that amplitude scale.

What the measurements here add is the size of the container: the total BBC step carries roughly eleven times the released margin in pairwise leverage, dominated by direct covariance alignment rather than parameter accommodation, with a probe-dependent sign. When the correction step is that much larger than the verdict, the verdict’s meaning rests entirely on the demonstrated smallness of the conditioned component — which existing validation supplies at the tested amplitude scale and not beyond it, and supplies by simulation rerun for FLRW-family forms rather than by direct test of non-FLRW functional shapes. The completion is specific: matched reruns with alternative laws as reference input, paired corrections against an empirically anchored selection model, and ultimately likelihood-level forward modelling. The paper asks for those measurements, not a presumption in either direction.

9 Conclusion

The DES-SN5YR v1.2 release carries a simulation-derived BBC correction with substantial redshift structure. Under the official redshift convention the released vector gives $\Delta\chi^2 = +1.1$ (Model B probe), the reconstructed pre-BBC vector $+12.8$; the pairwise discrimination leverage is $L_{\text{BBC}} = -11.7$, roughly eleven times the released discrimination, directed toward the probe, and unchanged when all 1829 entries are retained. Most of the leverage is direct covariance alignment ($L^{\text{fixed}} = -10.1$) rather than parameter accommodation (-1.6). Two further probes show the sensitivity is not unique to the discovering law, with probe-dependent sign and hence no consistent partisanship. The released-best-fit probe departs from ΛCDM by at most 0.055 mag offset-removed — well below

the ~ 0.15 mag reference-cosmology variation Camilleri et al. (2024) explicitly tested — so for this comparison prior DES simulation tests indicate the conditioned component of the leverage is small at this amplitude scale; the untested regime begins at larger departures — where our Einstein–de Sitter control (0.276 mag) shows two orders of magnitude larger leverage — and includes the pointwise shape-containment test and the direct coverage of non-FLRW functional forms. The quantitative requirement, Eq. (10), and the completion programme (matched reference-input reruns, paired corrections, likelihood-level forward modelling) define what remains to be measured before few- $\Delta\chi^2$ discrimination against distance laws far from the reference family can be considered conditioning-independent.

Data availability

All observational inputs are public DES-SN5YR v1.2 data products. The frozen analysis package will be archived under the reserved DOI 10.5281/zenodo.22022089 and is maintained at github.com/tracyphasespace/Model-Discrimination-DES-SN5YR, including the raw griz light curves for all 8,293 DES transient candidates, exact input provenance, the empirical sign test, all headline fits, robustness variants, projection diagnostics, the envelope comparison, and the systematic-mode propagation. No proprietary data were used.

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Author’s note on scope

This paper contains no proposed cosmological framework. The law of §5.2 is a probe used to test the sensitivity of pairwise model discrimination to a simulation-derived correction layer. A reader may reject that law physically without affecting the empirical characterization of `biasCor_mu`, the reconstruction of the pre-BBC vector, the envelope placement, or the completion programme defined above. A separate treatment of the propagation physics motivating the functional form of Model B is logically independent of this analysis.